

The missing L_1 endpoint in multilinear Fourier–Schur transference

Candidate full solution packet for arXiv:2206.00549

11 August 2026

Status. Candidate full affirmative answer, likely valid, pending specialist review. The result closes precisely the unimodular endpoint case isolated in Remark 3.3 of the source paper.

1 Source and exact question

The source is M. Caspers, A. Krishnaswamy-Usha, and G. Vos, *Multilinear transference of Fourier and Schur multipliers acting on non-commutative L_p -spaces*, arXiv:2206.00549v3, published in *Canadian Journal of Mathematics* 75 (2023), 1986–2006 [1]. The question occurs in Remark 3.3 on page 11 of the locally compiled arXiv source release.

MULTILINEAR TRANSFERENCE OF FOURIER AND SCHUR MULTIPLIERS

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Now, in equation (4.13) of [CJKM], the commutator terms vanish, as one of the terms in every case is a scalar. Additionally, since the $\widetilde{S}_j$ in the estimate for the first summand in equation (4.14) of [CJKM] is a scalar, we can once more avoid [CJKM Lemma 3.15] and the SAIN condition. Once again, note that since $x_j = 1$, the proof remains valid even when some of the $p_j = \infty$. $\square$

Remark 3.2. Theorem 3.1 assumes G to be second countable since its proof relies on [CJKM] which assumes second countability.

Remark 3.3. Fix $1 \leq i \leq n$. In case $p_i = p = 1$ and $p_j = \infty$ for all $1 \leq j \leq n, i \neq j$ we do not know whether Theorem 3.1 holds. The reason is that we do not know whether the limits (3.8) (at index i) hold and neither do we know if the two further applications of Proposition 3.9 in [CJKM Lemma 4.6, proof] hold.

4. AMENABLE GROUPS: TRANSFERENCE FROM SCHUR TO FOURIER MULTIPLIERS

Figure 1: The source question, Remark 3.3 on source PDF page 11.

Transcription. Fix $1 \leq i \leq n$. If $p_i = p = 1$ and $p_j = \infty$ for every $j \neq i$, does the Fourier-to-Schur transference theorem (Theorem 3.1 of [1]) still hold? The authors identify three unavailable endpoint uses: the limit in their equation (3.8) at index i , and two further uses of Proposition 3.9 inside the proof of CJKM Lemma 4.6 [2].

2 Definitions and statement

Let G be a locally compact, second-countable, unimodular group, let $\mathcal{L}(G)$ be its group von Neumann algebra, and let $L_q(\mathcal{L}(G))$ be the noncommutative L_q -space for the Plancherel trace.

Given $\phi \in C_b(G^n)$, put

$$\tilde{\phi}(s_0, \dots, s_n) = \phi(s_0 s_1^{-1}, s_1 s_2^{-1}, \dots, s_{n-1} s_n^{-1}).$$

The associated multilinear Fourier and Schur multipliers are denoted by T_ϕ and $M_{\tilde{\phi}}$, respectively. We use exactly the multiplicative amplifications and the $(p_1, \dots, p_n)$ -multiplicatively bounded norms of [1, Section 2].

Theorem 1 (Endpoint Fourier-to-Schur transference). *Let $1 \leq i \leq n$, set*

$$p = p_i = 1, \quad p_j = \infty \quad (j \neq i),$$

and suppose that $\phi \in C_b(G^n)$ is the symbol of a $(p_1, \dots, p_n)$ -multiplicatively bounded Fourier multiplier. Then $\tilde{\phi}$ is the symbol of a $(p_1, \dots, p_n)$ -multiplicatively bounded Schur multiplier, and

$$\|M_{\tilde{\phi}}\|_{(p_1, \dots, p_n)\text{-mb}} \leq \|T_\phi\|_{(p_1, \dots, p_n)\text{-mb}}.$$

Thus the question in Remark 3.3 of [1] has an affirmative answer.

3 Proof intuition

The interior-exponent proof in [1] concentrates the Plancherel mass near the identity by using $h_U^2 = k_U^* k_U$, where $k_U = \mu(U)^{-1/2} \lambda(1_U)$. Its only endpoint obstruction is an L_1 asymptotic-intertwining limit for Fourier-algebra multipliers.

At $p = 1$, taking a fractional power of h_U is unnecessary: the element h_U^2 already factors as the product of two normalized L_2 elements. Write a Fourier-algebra symbol as a coefficient $m(s) = \langle \pi(s)\xi, \eta \rangle$. Lift the two L_2 factors to Hilbert-space-valued columns. Their product is exactly the multiplier image. Because their Fourier supports lie in Ua^{-1} and Ub , strong continuity lets us freeze the representation vectors at a^{-1} and b . Noncommutative Hölder then makes the trace-norm error tend to zero. The frozen coefficient is $m(ab)$.

This one lemma replaces all three unavailable uses named by the source. In the two nested uses, every other exponent is infinity, so the intervening terms are scalars and all commutators vanish.

4 The endpoint coefficient lemma

We take Hilbert-space inner products linear in the first variable. We use the standard column space $L_2(\mathcal{L}(G); H_c)$: vector-valued Plancherel identifies $F \in L_2(G; H)$ with a column $\Lambda(F)$, with

$$\|\Lambda(F)\|_2^2 = \int_G \|F(s)\|_H^2 ds, \quad \|X^* Y\|_1 \leq \|X\|_2 \|Y\|_2. \quad (4.1)$$

Lemma 2 (Endpoint coefficient freezing). *Let U run through a neighborhood basis of the identity consisting of relatively compact measurable sets of positive measure. Set*

$$f_U = \mu(U)^{-1/2} 1_U, \quad k_U = \lambda(f_U) = u_U h_U.$$

For every $a, b \in G$ and every $m \in A(G)$,

$$\|T_m(\lambda_a h_U^2 \lambda_b) - m(ab) \lambda_a h_U^2 \lambda_b\|_1 \longrightarrow 0. \quad (4.2)$$

Consequently,

$$\|(T_m - \text{id})(\lambda_a h_U^2 \lambda_b)\|_1 \longrightarrow |m(ab) - 1|. \quad (4.3)$$

Proof. Choose a strongly continuous unitary representation $\pi : G \rightarrow \mathcal{U}(H)$ and $\xi, \eta \in H$ such that $m(s) = \langle \pi(s)\xi, \eta \rangle$. Put

$$A_U = k_U \lambda_{a^{-1}}, \quad B_U = k_U \lambda_b.$$

Then $A_U^* B_U = \lambda_a h_U^2 \lambda_b$ and $\|A_U\|_2 = \|B_U\|_2 = 1$.

Let $\alpha_U, \beta_U \in L_2(G)$ be the Fourier densities of A_U, B_U . Thus α_U is supported in Ua^{-1} and β_U in Ub . Define four vector-valued densities

$$\begin{aligned} P_U(s) &= \alpha_U(s) \pi(s) \eta, & P_U^0(s) &= \alpha_U(s) \pi(a^{-1}) \eta, \\ Q_U(t) &= \beta_U(t) \pi(t) \xi, & Q_U^0(t) &= \beta_U(t) \pi(b) \xi. \end{aligned} \tag{4.4}$$

For compactly supported scalar densities α, β , the Fourier density at r of $\Lambda(\alpha(\cdot) \pi(\cdot) \eta)^* \Lambda(\beta(\cdot) \pi(\cdot) \xi)$ is

$$\int_G \overline{\alpha(s)} \beta(sr) \langle \pi(sr) \xi, \pi(s) \eta \rangle ds = m(r) (\alpha^* * \beta)(r). \tag{4.5}$$

By L_2 approximation and (4.1), (4.5) remains valid here. Hence

$$\Lambda(P_U)^* \Lambda(Q_U) = T_m(A_U^* B_U). \tag{4.6}$$

The frozen product is

$$\Lambda(P_U^0)^* \Lambda(Q_U^0) = \langle \pi(b) \xi, \pi(a^{-1}) \eta \rangle A_U^* B_U = m(ab) A_U^* B_U. \tag{4.7}$$

Vector-valued Plancherel and the supports give

$$\begin{aligned} \|\Lambda(P_U) - \Lambda(P_U^0)\|_2 &\leq \sup_{u \in U} \|\pi(ua^{-1}) \eta - \pi(a^{-1}) \eta\|, \\ \|\Lambda(Q_U) - \Lambda(Q_U^0)\|_2 &\leq \sup_{v \in U} \|\pi(vb) \xi - \pi(b) \xi\|. \end{aligned} \tag{4.8}$$

Both quantities tend to zero by strong continuity. Applying (4.1) to the difference of (4.6) and (4.7) proves (4.2). Finally, $\|\lambda_a h_U^2 \lambda_b\|_1 = \text{Tr}(h_U^2) = \|k_U\|_2^2 = 1$, and the reverse triangle inequality applied to (4.2) proves (4.3). $\square$

Remark 3 (Quantitative form). *The proof bounds the left-hand side of (4.2) by*

$$\|\xi\| \sup_{u \in U} \|\pi(ua^{-1}) \eta - \pi(a^{-1}) \eta\| + \|\eta\| \sup_{v \in U} \|\pi(vb) \xi - \pi(b) \xi\|.$$

This makes the endpoint convergence independent of any L_1 operator norm for a symbol truncated to a small Fourier support.

5 Proof of Theorem 1

We follow the proof of [1, Theorem 3.1]; all references to equations in this section are to that paper. The authors already establish the argument when $1 < p_j \leq \infty$, and Remark 3.3 identifies exactly why their proof does not include the present vertex of the exponent simplex. We verify that Lemma 2 supplies every missing limit.

Fix a finite set $F \subset G$, a finite matrix amplification, and input matrices in the appropriate Schatten classes. The source introduces coordinatewise convolutions ϕ_k and the normalized elements $h_U^{2/q}$. The estimate through equations (3.4)–(3.7) uses only unitary invariance, Hölder's inequality, and the fact that coordinate convolution is norm decreasing. These statements remain valid at $q = 1, \infty$. The only issue is convergence of the error term.

The limits (3.8). The variables in (3.8) have the form

$$y_{j,U} = \lambda_{a_j} h_U^{2/p_j} \lambda_{b_j}.$$

When $j \neq i$, $p_j = \infty$, so $h_U^{2/p_j} = 1$ and the required identity is exact. At $j = i$, the symbol is a left-right translate of the compactly supported positive-definite function $\zeta \in A(G)$. It is therefore still in $A(G)$, and (4.3) is exactly the asserted limit (3.8). Dominated convergence then treats the source's C -term without change.

The two nested uses. For the B -term, [1] specializes the proof of [2, Lemma 4.6] to discrete inputs $x_j = 1$. As observed immediately before Remark 3.2 of [1], every commutator in CJKM equation (4.13) vanishes, and the later nested factor $\tilde{S}_j$ in equation (4.14) is a scalar. The first unavailable application of CJKM Proposition 3.9 is then

$$\|T_m(h_U^{2/p_n}) - m(e)h_U^{2/p_n}\|_{p_n} \longrightarrow 0. \quad (5.1)$$

It is exact if $p_n = \infty$, while if $i = n$ it is (4.2) with $a = b = e$.

In the telescoping part, the other unavailable expressions have the same form with a cumulative exponent q_j :

$$\|T_m(h_U^{2/q_j}) - m(e)h_U^{2/q_j}\|_{q_j} \longrightarrow 0. \quad (5.2)$$

Since precisely one reciprocal input exponent is nonzero, each q_j is either 1 or ∞ . The $q_j = \infty$ cases are exact and the $q_j = 1$ cases again follow from (4.2). Thus the nested B -term tends to zero. No fractional-power estimate or SAIN commutator estimate remains.

It follows that the source's finite-set estimate holds at the endpoint:

$$\|M_{\tilde{\phi}|_{F^{n+1}}}^{\sim}(a_1, \dots, a_n)\|_{S_1} \leq \|T_{\phi}\|_{(p_1, \dots, p_n)\text{-mb}} \prod_{j=1}^n \|a_j\|_{S_{p_j}}. \quad (5.3)$$

The same proof at every finite matrix amplification is unchanged: Lemma 2 tensors with matrix algebras, or equivalently is applied to the finitely many matrix coefficients. Finally the finite-truncation theorem [1, Theorem 2.2] turns (5.3) and its amplifications into the asserted global Schur-multiplier bound. This proves Theorem 1.

6 Verification, scope, and novelty

- The factorization was checked at the Fourier-density level in (4.5), so there is no appeal to the generally false principle that uniform smallness of a symbol controls its L_1 multiplier norm.
- Every nontrivial endpoint term in the source audit has exponent exactly one. Every other term has exponent infinity and is exact. This prevents an unproved interpolation or fractional-power step from entering the argument.
- The column proof is stable under finite matrix amplification, which is essential because the theorem concerns multiplicatively bounded norms.
- The result is limited to the locally compact, second-countable, unimodular setting of the source theorem. It does not claim the corresponding endpoint theorem for the non-unimodular framework of [3].

- Bounded novelty checks were performed on 11 August 2026: the run’s four lightweight indexes; exact-id, title, and endpoint-phrase searches across the locally ingested arXiv source corpus; and exact-id/endpoint searches on arXiv. The non-unimodular follow-up arXiv:2308.16595 explicitly repeats this endpoint as unknown in 2023 [3]. No later source found in these bounded searches claims the unimodular endpoint theorem. This is evidence, not an exhaustive literature guarantee.

Human-review recommendation. Specialist review is recommended. The highest-priority checks are (i) the column-product identity (4.5), including inner-product orientation, (ii) the claim that after the $x_j = 1$ specialization the two occurrences inside CJKM Lemma 4.6 are exactly (5.1)–(5.2), and (iii) the passage to all multiplicative amplifications.

References

- [1] M. Caspers, A. Krishnaswamy-Usha, and G. Vos, *Multilinear transference of Fourier and Schur multipliers acting on non-commutative L_p -spaces*, Canad. J. Math. 75 (2023), 1986–2006; arXiv:2206.00549.
- [2] M. Caspers, B. Janssens, A. Krishnaswamy-Usha, and L. Miaskiowski, *Local and multilinear noncommutative de Leeuw theorems*, Math. Ann. 388 (2024), 4251–4305; arXiv:2201.10400.
- [3] G. Vos, *Transference of multilinear Fourier and Schur multipliers acting on non-commutative L_p -spaces for non-unimodular groups*, arXiv:2308.16595.